

Web Engineering – Major Assignment Guidelines

Course: Web Engineering Lab – CSC-314(L)

Session: Fall 2025

Submission Format: ✓

Full project folder

✓ Database SQL file

✓ Documentation (PDF – includes SRS & thesis)

✓ PowerPoint presentation

🎯 Project Objective

To develop a **complete full-stack web application** with proper documentation and presentation, demonstrating skills learned in Web Engineering including frontend, backend, database integration, and SDLC practices.

◆ Technical Requirements

1□ Frontend

- Must use **HTML5, CSS3, Bootstrap**
- Responsive layout required
- Use **JavaScript** for interactivity and form validation

2□ Backend

- Must be built in **Core PHP (no frameworks)**
- CRUD operations required
- Use **MySQL** database

3□ Authentication

- User login system using **PHP Sessions**
- Must include **at least two user roles** (Admin + User OR similar)

4□ Modules

Your project must include at least **3 functional modules** depending on the assigned system

Example:

POS System → Product Module, Sales Module, Customer Module

School System → Student Module, Classes Module, Fees Module

5□ Database

- MySQL required
- Proper relational table structure
- Submit **.sql export file**

6□ Validation

- Client-side (JavaScript)
- Server-side (PHP)

7□ Design Requirement

- Must be clean, responsive, user-friendly
 - Use appropriate Bootstrap components such as:
 - Navbar
 - Tables
 - Forms
 - Buttons
 - Modals
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Documentation Requirements

1□ Requirements Gathering

Every student must collect requirements for their assigned project, including:

- Stakeholders
 - Objectives
 - Features
 - Functional + non-functional requirements
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2 SRS (Software Requirements Specification)

This is compulsory.

Must follow IEEE format & include:

- Introduction & System Overview
- Problem Statement
- Scope
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions + Diagram
- System Flow / DFD / Architecture
- UI Wireframes (if applicable)
- Database requirements / ERD

✦ Final SRS will be submitted with documentation in theory class.

3 ☐ System Design Documentation

Required:

- ✓ ERD
 - ✓ Database Schema
 - ✓ Module Descriptions
 - ✓ UI Layout Structure
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4 ☐ Project Thesis / Final Report

The final report must include:

- Abstract
- Introduction
- Problem definition
- Objectives
- Literature / Existing systems (optional)
- Methodology
- Requirements
- Design
- Implementation
- Screenshots
- Testing

- Results & Conclusion
- Future scope
- References

✦ Thesis must be submitted in theory class.

Presentation Requirements

Each student must prepare and present a **PowerPoint presentation** including:

- Title & information
 - Problem statement
 - Requirements overview
 - Tech stack
 - Screenshots/demo
 - Results & conclusion
 - Time limit: **10 minutes max**
 - ✦ Must be presented individually
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Important Rules & Notes

✗ Plagiarism = ZERO MARKS

- No copying from other students
- No duplicate projects

✓ Allowed

- External libraries for UI
- Template-based UI only if backend is fully custom
- GitHub for inspiration only (not copy-paste)
- Prompt Engineering

Submission Instructions

- ✓ Submit full project folder (Lab)
- ✓ Documentation PDF (SRS + Thesis) (Theory)
- ✓ ✦ **Submit via LMS on assigned date.**